

Search and Matching: McCall and Diamond–Mortensen–Pissarides

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Fall 2026

From Accounting to Theory

- The lake model established that unemployment is governed by two hazards: the separation rate s_t and the job-finding rate f_t
- In the data, the cyclical action is mostly in f_t .
- This topic builds the theory that makes those hazards **choices and equilibrium objects**.

Two Building Blocks

1. **McCall (1970):** one worker's search problem.

- Makes the *acceptance* decision economic;
- the offer arrival is still exogenous.

2. **DMP:** the two-sided general equilibrium.

- Firms create the offers;
- the matching function converts vacancies and searchers into meetings;
- wages split the surplus.

Payoffs

- A theory of the job-finding probability $\lambda_w(\theta)$.
- A theory of wages under bilateral monopoly.
- A welfare benchmark (Hosios).

Environment

- Discrete time; a single risk-neutral, infinitely-lived worker; discount factor $\beta \in (0, 1)$.
- While unemployed: income b (unemployment benefit, home production, leisure value) and one wage offer w per period, drawn i.i.d. from a known distribution $F(w)$ on $[0, \bar{w}]$.
- Decision: **accept** (work at w forever) or **reject** (get b , draw again next period).
 - No recall for now (irrelevant here: the problem is stationary).
- Everything about the labor market is packed into F and the one-offer-per-period arrival: partial equilibrium on purpose.

Search as Optimal Stopping

- Search is an **optimal stopping problem**.
- The worker weighs a sure thing today against the option of a better draw tomorrow.

Value Functions

Value of holding wage w forever:

$$W(w) = \frac{w}{1 - \beta}.$$

Value of being unemployed (before this period's draw is accepted or rejected):

$$U = b + \beta \int_0^{\bar{w}} \max \left\{ \frac{w'}{1 - \beta}, U \right\} dF(w').$$

Where the Economics Lives

- The max operator is where all the economics lives: the worker keeps the *option* to reject.
- The map defining U is a contraction (Blackwell), so U exists and is unique.
 - Iterate to compute it.

The Reservation Wage

Since $W(w)$ is increasing and U is a constant, the optimal policy is a cutoff rule:

$$\text{accept} \iff w \geq w^*, \quad \text{where } \frac{w^*}{1 - \beta} = U.$$

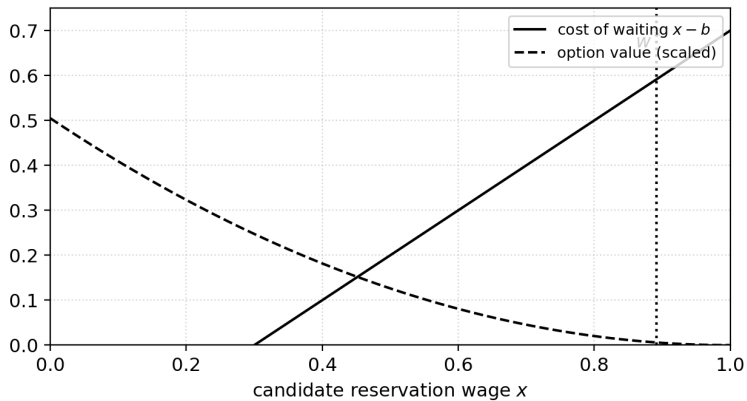
Substituting and rearranging yields the **reservation wage equation**:

$$w^* - b = \frac{\beta}{1 - \beta} \int_{w^*}^{\bar{w}} (w - w^*) dF(w).$$

Reading the Reservation Wage Equation

- Left-hand side: the *cost of rejecting* a wage exactly at w^* this period (forgone earnings over b).
- Right-hand side: the *option value of continued search*, the discounted expected gain from drawing something better than w^* .
- At the reservation wage, the worker is exactly indifferent: marginal cost of waiting = marginal benefit.

The Reservation Wage, Graphically



Reading the Figure

- The cost of waiting is increasing in x .
- The option value is decreasing in x : fewer better draws remain.
- A unique crossing pins down w^* .

Comparative Statics

From the reservation wage equation:

- **Benefits.** $\uparrow b \Rightarrow \uparrow w^*$: search is subsidized, workers hold out for more. Unemployment duration rises.
 - The moral-hazard margin of UI; welfare analysis needs risk aversion.
- **Patience.** $\uparrow \beta \Rightarrow \uparrow w^*$: the option value is a claim on future draws.
- **Dispersion.** A **mean-preserving spread** of F raises w^* .
 - The max operator makes the worker's payoff convex in w : bad draws are rejected anyway, good draws are kept.
 - *Search loves risk.*

Worked Example F uniform on $[0, 1]$, $\beta = 0.99$:

	$b = 0.3$	$b = 0.6$
Reservation wage w^*	0.891	0.920
Job-finding probability $\lambda = 1 - F(w^*)$	0.109	0.080
Mean duration (periods)	9.2	12.4

From Policy to Hazard

- The model's job-finding probability is $\lambda = 1 - F(w^*)$.
- Unemployment duration is geometric with mean $1/\lambda$.
- Search theory converts *primitives* (b, β, F) into the hazard that we measured during the first week.

What Is Missing

- Where does F come from? Firms are absent.
- **The Diamond (1971) paradox:** if identical firms *post* wages and workers must pay any search cost $\varepsilon > 0$, the unique equilibrium has *all* firms posting the monopsony wage.
 - Wage dispersion collapses.
 - Sequential search theory alone cannot support the distribution it assumes.
- Two escape routes:
 - bilateral *bargaining* over match surplus (DMP, next);
 - on-the-job search generating dispersion (Burdett–Mortensen; a future field course).

Environment

Discrete time (a month); risk-neutral workers and firms; common discount factor $\beta = 1/(1 + r)$.

- Unit mass of workers: employed (produce z , earn w) or unemployed (get $b < z$, search).
- Free mass of potential firms; a vacancy costs κ per period; a filled job produces z .
- Matches dissolve with probability σ each period (endogenized later).
- Wages: **generalized Nash bargaining** with worker weight $\gamma \in (0, 1)$, renegotiated every period.

Notation Map

Object	These slides / AKMM
discount	$\beta (= 1/(1 + r))$
separation probability	σ
job-finding probability	$\lambda_w(\theta)$
vacancy-filling probability	$\lambda_f(\theta)$
matching function	$\chi u^\eta v^{1-\eta}$
vacancy cost	κ
productivity	z
worker bargaining weight	γ

Meetings and Unemployment

- **Meetings** via the matching function $M(u, v) = \chi u^\eta v^{1-\eta}$, CRS, so with tightness $\theta = v/u$,

$$\lambda_w(\theta) = \frac{M(u, v)}{u} = \chi \theta^{1-\eta}, \quad \lambda_f(\theta) = \frac{M(u, v)}{v} = \chi \theta^{-\eta}, \quad \lambda_w = \theta \lambda_f.$$

- **Timing:** matches formed in t produce from $t + 1$; separations hit at the end of t .
- Unemployment evolves as in the lake model:

$$u_{t+1} = (1 - \lambda_w(\theta_t)) u_t + \sigma(1 - u_t), \quad \text{in steady state } u = \frac{\sigma}{\sigma + \lambda_w(\theta)}.$$

- The model's job is to determine θ .

Firm Values

Let J be the value of a filled job and V the value of a vacancy. Bellman equations in steady state:

$$J = z - w + \beta[(1 - \sigma)J + \sigma V], \quad V = -\kappa + \beta[\lambda_f(\theta)J + (1 - \lambda_f(\theta))V].$$

- Flow payoff today plus discounted continuation: a filled job earns $z - w$ and survives with probability $1 - \sigma$; a vacancy pays κ and is filled with probability $\lambda_f(\theta)$, producing from next period.

Free Entry and the Job Creation Condition

- **Free entry** drives $V = 0$: vacancies are posted until the marginal one breaks even. Then

$$\frac{\kappa}{\lambda_f(\theta)} = \beta J = \frac{\beta(z - w)}{1 - \beta(1 - \sigma)} = \frac{z - w}{r + \sigma},$$

using $(1 - \beta(1 - \sigma))/\beta = r + \sigma$.

- Interpretation: expected recruiting cost per hire (κ per period for an expected duration $1/\lambda_f(\theta)$) equals the discounted value of the filled job.
- **This is the job creation condition.** It slopes *down* in (θ, w) space: higher wages support fewer vacancies per searcher.

Worker Values and the Surplus

Let W be the value of employment and U the value of unemployment:

$$W = w + \beta[(1 - \sigma)W + \sigma U], \quad U = b + \beta[\lambda_w(\theta)W + (1 - \lambda_w(\theta))U].$$

Define the **match surplus**

$$S = (W - U) + (J - V) = (W - U) + J \quad (\text{using } V = 0),$$

the joint gain from the match over the parties' outside options.

Nash Bargaining

The wage maximizes $(W - U)\gamma(J - V)^{1-\gamma}$. Both surpluses are linear in w (slopes $+1, -1$), so the first-order condition is a fixed split:

$$(1 - \gamma)(W - U) = \gamma J \iff W - U = \gamma S, \quad J = (1 - \gamma) S.$$

- Why bargaining? A meeting creates a *bilateral monopoly*: the match surplus exists only inside the pair, so the law of one price has no bite.
- Frictions make wage determination a genuine modeling choice.

The Surplus Equation

Add the worker's and the firm's equations, and use $W - U = \gamma S$ in the job-finding term:

$$S = (z - b) + \beta(1 - \sigma - \gamma\lambda_w(\theta))S \implies S = \frac{z - b}{1 - \beta(1 - \sigma) + \beta\gamma\lambda_w(\theta)}.$$

- The wage has dropped out: it is a transfer inside the match.
- The surplus is eroded by separation (σ) and by the worker's improving outside option ($\gamma\lambda_w(\theta)$): a tighter market makes unemployment less costly.
- Free entry in surplus form: $\kappa = \lambda_f(\theta)(1 - \gamma)\beta S$.

The Wage Equation

Combining surplus splitting with the value equations and free entry:

$$w = (1 - \gamma) b + \gamma (z + \kappa \theta).$$

- A weighted average of the outside option b and the firm's full benefit from the match: output z *plus* saved recruiting costs $\kappa \theta = \kappa v / u$.
 - With one more worker employed, the market is tighter for the remaining firms; the worker captures a share of that.
- Identical to the continuous-time wage equation: timing conventions do not touch it.

Reading the Wage Equation

- Wages *increase in tightness*: this is the model's wage cyclicalilty.
- Higher z raises θ , which raises w , which claws back most of the incentive to post vacancies.
- Slopes *up* in (θ, w) space: the wage curve.

Equilibrium

Two equations in (θ, w) :

$$w = z - \frac{(r + \sigma) \kappa}{\lambda_f(\theta)} \quad (\text{JC})$$

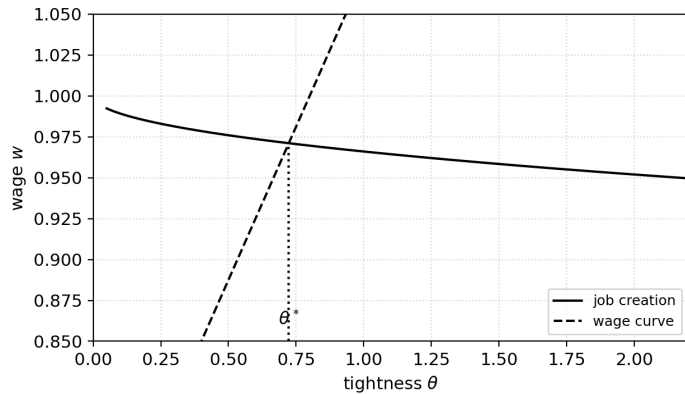
$$w = (1 - \gamma) b + \gamma(z + \kappa\theta) \quad (\text{WC})$$

Eliminating w : one equation in θ ,

$$\frac{(r + \sigma) \kappa}{\lambda_f(\theta)} + \gamma \kappa \theta = (1 - \gamma)(z - b) \quad (\text{AKMM 20.17, rearranged}).$$

- Left side increasing from 0, right side a positive constant: a unique θ^* whenever $z > b$.
- Then $u^* = \sigma / (\sigma + \lambda_w(\theta^*))$ and $v^* = \theta^* u^*$: the model **closes** on the Beveridge curve (AKMM, Figure 20.5).

Equilibrium, Graphically



Worked Example: A Monthly Calibration

Parameter	Value	Target / source
β	0.996	$r = 0.004$, about 5% annual rate
σ	0.02	separation probability
η	0.5	Petrongolo–Pissarides range
γ	0.5	symmetric bargaining (and Hosios: $\gamma = \eta$)
z	1	normalization
b	0.4	Shimer's (2005) value
χ	0.53	to hit $\lambda_w = 0.45$
κ	0.75	to hit $\theta \approx 0.72$

Steady state: $\theta^* = 0.723$, $\lambda_w = 0.451$, $\lambda_f = 0.623$ (mean vacancy duration 1.6 months), $w = 0.971$,
 $u^* = 4.25\%$.

The Textbook's Calibration

AKMM (Table 20.1): $\beta = 0.996$, $\sigma = 0.034$, $\chi = 0.45$, $b = 0.4$, $\gamma = \eta = 0.72$, $\bar{\theta} = 1$, κ backed out from the job creation condition.

- Gives $\kappa = 0.21$ and $u = 7.0\%$.
- Higher σ , lower χ : Shimer's inflow/outflow rates include movements through non-participation.
- $\eta = 0.72$ is Shimer's matching-function estimate.
- Same questions, same answers; ours is tied to the flows we measured in our first week.

		Baseline	$z : 1 \rightarrow 1.02$	$b : 0.4 \rightarrow 0.7$
Comparative Statics	Tightness θ	0.723	0.748	0.347
	Job-finding λ_w	0.451	0.458	0.312
	Unemployment u	4.25%	4.18%	6.02%

Elasticity of tightness with respect to productivity:

$$\varepsilon_{\theta z} = \frac{d \ln \theta}{d \ln z} = \frac{z}{\theta} \cdot \frac{1 - \gamma}{G'(\theta)}, \quad G(\theta) = \frac{(r + \sigma)\kappa}{\chi} \theta^\eta + \gamma \kappa \theta,$$

equal to 1.75 at the baseline; in the limit $r, \gamma \rightarrow 0$, $\varepsilon_{\theta z} = \frac{1}{\eta} \cdot \frac{z}{z - b}$.

Preview of Shimer Puzzle

- Productivity up 2 percent $\Rightarrow$ θ up 3.5 percent, u down by **0.07 percentage points**.
- U.S. unemployment is about 10 times as volatile as productivity. The baseline model delivers a small fraction of that.
 - *That gap is the Shimer puzzle.*
 - To get $\varepsilon_{\theta z} = 20$ one needs $b = 0.94$: the Hagedorn–Manovskii debate, in one number.
- Note also the huge sensitivity to b : raising the outside option shrinks the surplus $z - b$ and moves u a lot.
 - The two observations are the same fact in disguise (fundamental surplus, Ljungqvist and Sargent, Ch. 30).

Is Equilibrium Unemployment Efficient?

Search creates two externalities that private agents do not price:

- **Congestion:** an extra vacancy lowers every other firm's filling probability $\lambda_f(\theta)$; an extra searcher lowers other workers' $\lambda_w(\theta)$.
- **Thick market:** the same entries *raise* the meeting probability on the other side of the market.

The Planner's Problem

Choose vacancies to maximize output net of vacancy costs, subject to the matching technology:

$$\max_{\{v_t\}} \sum_t \beta^t [z(1 - u_t) + b u_t - \kappa v_t] \quad \text{s.t.} \quad u_{t+1} = u_t - M(u_t, v_t) + \sigma(1 - u_t).$$

Steady-state optimality condition :

$$\frac{(r + \sigma)\kappa}{\lambda_f(\theta)} + \eta\kappa\theta = (1 - \eta)(z - b).$$

- Compare with the equilibrium condition: identical with γ replaced by η .

The Hosios (1990) Condition

The decentralized equilibrium coincides with the planner's solution if and only if

$$\gamma = \eta,$$

the worker's bargaining share equals the elasticity of matching with respect to unemployment.

- Intuition (AKMM §20.4.3): a firm values a vacancy at its *average* product λ_f times its share $1 - \gamma$; the planner at its *marginal* product $(1 - \eta)\lambda_f$ times the whole surplus. Congestion and surplus-sharing offset exactly when $\gamma = \eta$.
 - $\gamma > \eta$: too little entry, u too high.
 - $\gamma < \eta$: vacancy congestion, too much entry.
- No market force aligns γ with η : a knife edge; policy (UI, hiring subsidies) has a potential corrective role.

Dynamics out of Steady State

With time subscripts, free entry and Nash bargaining hold *every period*:

$$\frac{\kappa}{\lambda_f(\theta_t)} = \beta J_{t+1}, \quad (1 - \gamma)(W_t - U_t) = \gamma J_t.$$

Define $S_t \equiv (W_{t+1} - U_{t+1}) + J_{t+1}$, the surplus of a match in place next period. Then (AKMM 20.18–20.19)

$$\kappa = \lambda_f(\theta_t)(1 - \gamma)\beta S_t, \quad S_t = (z - b) + \beta(1 - \sigma - \gamma\lambda_w(\theta_{t+1}))S_{t+1}.$$

- A difference equation in θ_t alone: u_t appears nowhere.
- The wage equation still holds period by period: $w_t = (1 - \gamma)b + \gamma(z + \kappa\theta_t)$.

Tightness Is a Jump Variable

Write $x_t \equiv \beta S_t = \kappa \theta_t^\eta / [(1 - \gamma)\chi]$ (increasing in θ_t). The equilibrium map is

$$x_t = \beta(z - b) + \beta(1 - \sigma - \gamma\lambda_w(\theta_{t+1})) x_{t+1} \quad (\text{AKMM 20.16}).$$

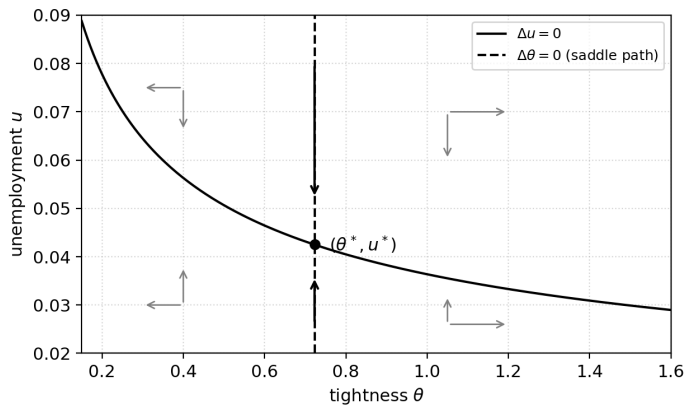
- If $(1 - \sigma)\eta > \gamma\lambda_w(\theta)$ (true at monthly frequency), this defines $\theta_{t+1} = g(\theta_t)$ with

$$g'(\theta^*) = \frac{\eta}{\beta[(1 - \sigma)\eta - \gamma\lambda_w(\theta^*)]} > 1 \quad (= 1.90 \text{ at the baseline}).$$

- The steady state is **unstable** in θ : any path off θ^* explodes or hits zero, violating free entry.
- So $\theta_t = \theta^*$ for all t , regardless of u_0 , and

$$u_t = u^* + (u_0 - u^*)(1 - \sigma - \lambda_w(\theta^*))^t,$$

The Phase Diagram



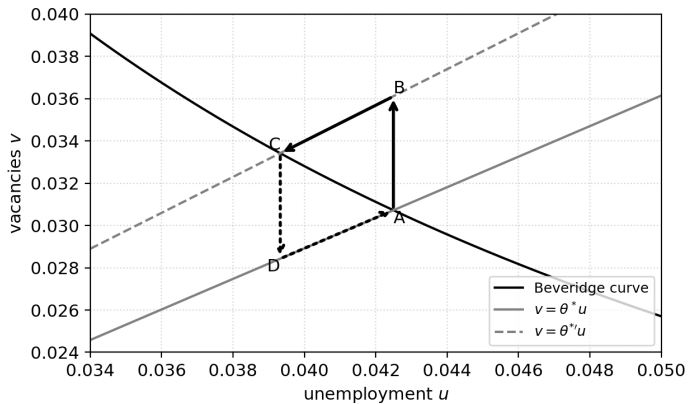
The $\Delta u = 0$ locus is the Beveridge relation; the $\Delta \theta = 0$ locus is the vertical line $\theta = \theta^*$, which is also the saddle path. Grey arrows: motion off the saddle path.

Permanent Shocks and the Beveridge Loop

Let z rise permanently at $t = 0$.

- θ jumps at once to the new θ^{*} ; u declines gradually.
- Vacancies $v_t = \theta^{*} u_t$ **overshoot**: up on impact (to $\theta^{*} u^*$), then down along the new ray as u falls.
- At the baseline, $z : 1 \rightarrow 1.02$: v jumps 3.5%, settles 1.8% higher; u falls from 4.25% to 4.18%, half of it in the first month.
- Reverse the shock and the (u, v) path closes a **counterclockwise loop**: Topic 1's loops as an equilibrium phenomenon.

The Beveridge Loop, Graphically



Permanent rise in z from 1 to 1.10 ($\theta^* : 0.723 \rightarrow 0.850$). Solid: $A \rightarrow B$ on impact, then down the new ray to C . Dotted: the return path if the shock is reversed.

Endogenizing Separations (Mortensen–Pissarides 1994)

- Each match has an idiosyncratic productivity $x \in [0, 1]$ and produces zx ; new matches start at $x = 1$; each period, with probability π , a new x is drawn from G .
- Nash bargaining at every x : $W(x) - U = \gamma S(x)$, $J(x) = (1 - \gamma)S(x)$. Both sides *agree* on when to separate: separations are privately efficient.
- The surplus is linear in x , so the rule is a threshold: continue iff $x \geq R$, the **reservation productivity**, $S(R) = 0$. Separations become $\sigma = \pi G(R)$.

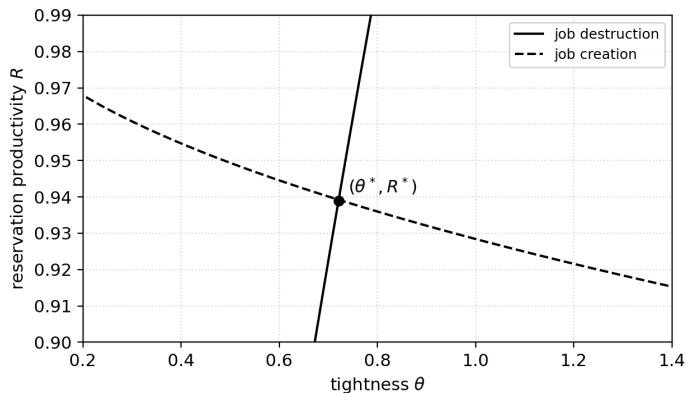
Job Destruction and Job Creation

$$R = \frac{1}{z} \left(b + \frac{\gamma}{1-\gamma} \kappa \theta \right) - \frac{\pi}{r + \pi} \int_R^1 (x' - R) dG(x') \quad (\text{JD})$$

$$\frac{\kappa}{\lambda_f(\theta)} = (1 - \gamma) \frac{z(1 - R)}{r + \pi} \quad (\text{JC})$$

- JD: outside option relative to output, minus an **option value**: a match at R runs a loss today for the chance of a better draw. R rises with b and with θ .
- JC: the value of a new job is the distance $1 - R$ to the exit margin, capitalized at $r + \pi$.
- JD slopes up, JC slopes down in (θ, R) : a unique intersection; u enters neither.
- Wages: $w(x) = (1 - \gamma)b + \gamma(zx + \kappa\theta)$; at $x = R$ the firm pays more than the match produces.

Job Destruction and Job Creation, Graphically



G uniform on $[0, 1]$, $\pi = 0.0213$, all other parameters as in the baseline: $(\theta^*, R^*) = (0.72, 0.94)$, $\sigma = 0.02$, $u = 4.26\%$. With this G only the top few percent of draws are worth keeping.

Worked Example: A 2% Recession

	Baseline	$z = 0.98$	$z = 0.98, \theta$ fixed	$b = 0.7$
R	0.939	0.939	0.959	0.958
θ	0.721	0.696	0.721	0.345
$\sigma = \pi G(R)$	0.0200	0.0200	0.0204	0.0204
u	4.26%	4.33%	4.34%	6.15%

- Partial equilibrium (θ fixed): R jumps two points, 2% of matches destroyed at once.
- General equilibrium: θ falls, the outside option $b + \frac{\gamma}{1-\gamma}\kappa\theta$ falls with it, and R does not move at all.

When Is Destruction Countercyclical?

Substituting JC into JD, R falls when z rises if and only if

$$b > \frac{1-\eta}{\eta} \cdot \frac{\gamma}{1-\gamma} \kappa\theta \quad (\text{under Hosios: } b > \kappa\theta).$$

- The outside option has a rigid part, b , and a flexible part, $\kappa\theta$, which falls faster than z in a recession (elasticity $1/\eta$ along JC).
- Destruction is countercyclical only if the rigid part dominates: at the baseline $0.40 < 0.54$ (fails); at $b = 0.7$ it holds, but the spike is still tiny.
- **Same lesson as Topic 3:** the flexibility of the Nash outside option mutes both margins. A rigid outside option revives both.

The Destruction Spike

- R_t and θ_t are both jump variables; the states are u and the distribution of x across matches (a mass $G(R)$ of employment at $x = 1$, density $(1 - u)g(x)$ on $[R, 1)$).
- When R jumps to R' , every match with $x \in [R, R')$ dies *at once*: a fraction $G(R') - G(R)$ of employment.
- Creation cannot spike: λ_w works on the stock of unemployed, which builds up slowly. Inflows lead at recession onsets, as in Topic 1.
- Caveat (Shimer 2005; Mortensen–Nagypál 2007): separation shocks raise u fast and can push $v = \theta u$ up on impact, against the Beveridge curve.

Endogenous Separations: What They Deliver and Cost

- Deliver: a theory of the separation flow, the timing of inflows, and the effect of b or firing costs on job loss.
- Cost: an extra fixed point (the (R, θ) pair), and with Nash wages little extra cyclical action.
- AKMM (§20.7) endogenize σ differently: the firm chooses σ at a maintenance cost $c(\sigma) = \phi\sigma^{-\xi}$, with $-c'(\sigma) = \kappa/\lambda_f(\theta)$.
 - Nash wages: u still far too smooth (Table 20.5).
 - Fixed wages: endogenous terms cancel in the log-linear JC condition; u now *over-shoots* the data (Table 20.6). The rigidity, not the device, is what matters.
- The constant- σ model remains the workhorse, consistent with the ins-and-outs evidence (separations: a third to a half of the variance, concentrated at onsets).

Computing and Simulating the Model

- **Steady state:** one nonlinear equation in θ ,

$$\frac{(r + \sigma) \kappa}{\lambda_f(\theta)} + \gamma \kappa \theta = (1 - \gamma)(z - b),$$

solved by bisection in one line. Everything else is arithmetic. If your numbers are off, suspect timing (a missing β), not rounding.

- **Stochastic version:** In z_t an AR(1) ($\rho = 0.949$, s.d. 0.0065 monthly, AKMM Table 20.1).
 - Log-linearize and read off C (AKMM §20.6.1); or
 - discretize (Rouwenhorst) and iterate on
$$S(z_i) = \sum_j \Pi_{ij} [z_j - b + \beta(1 - \sigma - \gamma \lambda_w(\theta_j)) S(z_j)], \quad \kappa / \lambda_f(\theta_i) = \beta(1 - \gamma) S(z_i);$$
 - simulate, aggregate to quarters, HP-filter (1,600), and compare with AKMM Tables 20 2–20 3

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